

# INTERNATIONAL STANDARD

---

**Automatic electrical controls –  
Part 2-13: Particular requirements for humidity sensing controls**



## **THIS PUBLICATION IS COPYRIGHT PROTECTED**

**Copyright © 2017 IEC, Geneva, Switzerland**

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office  
3, rue de Varembe  
CH-1211 Geneva 20  
Switzerland

Tel.: +41 22 919 02 11  
Fax: +41 22 919 03 00  
[info@iec.ch](mailto:info@iec.ch)  
[www.iec.ch](http://www.iec.ch)

### **About the IEC**

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

### **About IEC publications**

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

#### **IEC Catalogue - [webstore.iec.ch/catalogue](http://webstore.iec.ch/catalogue)**

The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

#### **IEC publications search - [www.iec.ch/searchpub](http://www.iec.ch/searchpub)**

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, replaced and withdrawn publications.

#### **IEC Just Published - [webstore.iec.ch/justpublished](http://webstore.iec.ch/justpublished)**

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

#### **Electropedia - [www.electropedia.org](http://www.electropedia.org)**

The world's leading online dictionary of electronic and electrical terms containing 20 000 terms and definitions in English and French, with equivalent terms in 16 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

#### **IEC Glossary - [std.iec.ch/glossary](http://std.iec.ch/glossary)**

65 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

#### **IEC Customer Service Centre - [webstore.iec.ch/csc](http://webstore.iec.ch/csc)**

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: [csc@iec.ch](mailto:csc@iec.ch).



IEC 60730-2-13

Edition 3.0 2017-10

# INTERNATIONAL STANDARD

---

**Automatic electrical controls –  
Part 2-13: Particular requirements for humidity sensing controls**

INTERNATIONAL  
ELECTROTECHNICAL  
COMMISSION

ICS 97.120

ISBN 978-2-8322-4900-0

**Warning! Make sure that you obtained this publication from an authorized distributor.**

## CONTENTS

|                                                                                                                 |    |
|-----------------------------------------------------------------------------------------------------------------|----|
| FOREWORD.....                                                                                                   | 3  |
| 1 Scope and normative references .....                                                                          | 5  |
| 2 Terms and definitions .....                                                                                   | 5  |
| 3 General requirements .....                                                                                    | 6  |
| 4 General notes on tests .....                                                                                  | 6  |
| 5 Rating.....                                                                                                   | 6  |
| 6 Classification.....                                                                                           | 6  |
| 7 Information .....                                                                                             | 6  |
| 8 Protection against electric shock .....                                                                       | 6  |
| 9 Provision for protective earthing .....                                                                       | 6  |
| 10 Terminals and terminations.....                                                                              | 6  |
| 11 Constructional requirements .....                                                                            | 6  |
| 12 Moisture and dust resistance .....                                                                           | 6  |
| 13 Electric strength and insulation resistance .....                                                            | 7  |
| 14 Heating.....                                                                                                 | 7  |
| 15 Manufacturing deviation and drift.....                                                                       | 7  |
| 16 Environmental stress .....                                                                                   | 7  |
| 17 Endurance .....                                                                                              | 7  |
| 18 Mechanical strength .....                                                                                    | 8  |
| 19 Threaded parts and connections.....                                                                          | 8  |
| 20 Creepage distances, clearances and distances through solid insulation.....                                   | 8  |
| 21 Resistance to heat, fire and tracking.....                                                                   | 8  |
| 22 Resistance to corrosion .....                                                                                | 8  |
| 23 Electromagnetic compatibility (EMC) requirements – Emission .....                                            | 8  |
| 24 Components .....                                                                                             | 8  |
| 25 Normal operation .....                                                                                       | 9  |
| 26 Electromagnetic compatibility (EMC) requirements – Immunity .....                                            | 9  |
| 27 Abnormal operation .....                                                                                     | 9  |
| 28 Guidance on the use of electronic disconnection .....                                                        | 9  |
| Annex H (normative) Requirements for electronic controls .....                                                  | 9  |
| Annex AA (normative) Independently mounted and in-line cord controls .....                                      | 16 |
| Annex BB (normative) Regional differences .....                                                                 | 17 |
| Annex CC (informative) Specific regional requirements in Japan.....                                             | 19 |
| Bibliography.....                                                                                               | 20 |
| Table H.101 – Compliance criteria .....                                                                         | 11 |
| Table AA.1 – Number of cycles .....                                                                             | 16 |
| Table BB.1 – Minimum number of cycles for independently mounted and in-line cord controls (United States) ..... | 17 |
| Table BB.2 – Minimum number of cycles for independently mounted and in-line cord controls (Canada) .....        | 18 |

## INTERNATIONAL ELECTROTECHNICAL COMMISSION

**AUTOMATIC ELECTRICAL CONTROLS –****Part 2-13: Particular requirements for humidity sensing controls****FOREWORD**

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as “IEC Publication(s)”). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 60730-2-13 has been prepared by IEC technical committee 72: Automatic electrical controls.

The text of this standard is based on the following documents:

| FDIS         | Report on voting |
|--------------|------------------|
| 72/1078/FDIS | 72/1108/RVD      |

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

This third edition cancels and replaces the second edition published in 2006. This edition constitutes a technical revision. This edition includes alignment with the text of 60730-1 fifth edition and the following significant technical changes with respect to the previous edition:

- a) alignment of the EMC requirements in Clause H.26 to those in other part 2 standards;
- b) addition of requirements in Clause H.27 to cover class B and C control functions of humidity sensing controls.

This Part 2-13 is intended to be used in conjunction with IEC 60730-1. It was established on the basis of the fifth edition of that standard (2013). Consideration may be given to future editions of, or amendments to, IEC 60730-1.

This Part 2-13 supplements or modifies the corresponding clauses in IEC 60730-1, so as to convert that publication into the IEC standard: Particular requirements for humidity sensing controls.

Where this Part 2-13 states "addition", "modification" or "replacement", the relevant requirement, test specification or explanatory matter in Part 1 should be adapted accordingly.

Where no change is necessary, this Part 2-13 indicates that the relevant clause or subclause applies.

In the development of a fully international standard it has been necessary to take into consideration the differing requirements resulting from practical experience in various parts of the world and to recognize the variation in national electrical systems and wiring rules.

In this publication, the following print types are used:

- Requirements proper: in roman type.
- *Test specifications: in italic type.*
- Explanatory matter: in smaller roman type.

Subclauses, notes or items which are additional to those in Part 1 are numbered starting from 101, additional annexes are lettered AA, BB, etc.

A list of all parts of the IEC 60730 series, under the general title *Automatic electrical controls* can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

## **AUTOMATIC ELECTRICAL CONTROLS –**

### **Part 2-13: Particular requirements for humidity sensing controls**

#### **1 Scope and normative references**

This clause of Part 1 is applicable except as follows:

##### **1.1 Scope**

*Replacement:*

This part of IEC 60730 applies to automatic electrical humidity sensing controls for use in, on or in association with equipment, including controls for heating, air-conditioning and similar applications. The equipment may use electricity, gas, oil, solid fuel, solar thermal energy, etc. or a combination thereof.

NOTE Throughout this standard, the word "equipment" includes "appliance" and "control system".

This International Standard is applicable to automatic electrical humidity sensing controls forming part of a building automation control system within the scope of ISO 16484.

This standard also applies to automatic electrical humidity sensing controls for equipment that may be used by the public, such as equipment intended to be used in shops, offices, hospitals, farms and commercial and industrial applications.

This standard does not apply to automatic electrical humidity sensing controls intended exclusively for industrial process applications unless explicitly mentioned in the equipment standard.

##### **1.1.2 Replacement:**

This standard applies to automatic electrical controls, mechanically or electrically operated, responsive to or controlling humidity.

##### **1.1.3 Not applicable.**

#### **2 Terms and definitions**

This clause of Part 1 is applicable except as follows:

##### **2.2 Definitions of types of control according to purpose**

###### **2.2.19 Addition:**

Note 1 to entry: In general, a **humidity sensing control** is an **operating control**.

*Additional definitions:*

###### **2.2.101**

###### **humidity sensing control**

automatic **electrical control** which is either intended to keep the controlled humidity above, below or between a particular value(s)

### **2.2.102**

#### **room humidistat**

independently mounted or incorporated **humidity sensing control** intended to control the humidity of habitable space

## **3 General requirements**

This clause of Part 1 is applicable.

## **4 General notes on tests**

This clause of Part 1 is applicable.

## **5 Rating**

This clause of Part 1 is applicable.

## **6 Classification**

This clause of Part 1 is applicable except as follows:

### **6.3.9** *Additional subclauses:*

**6.3.9.101 – humidity sensing control;**

**6.3.9.102 – room humidistat;**

## **7 Information**

This clause of Part 1 is applicable.

## **8 Protection against electric shock**

This clause of Part 1 is applicable.

## **9 Provision for protective earthing**

This clause of Part 1 is applicable.

## **10 Terminals and terminations**

This clause of Part 1 is applicable.

## **11 Constructional requirements**

This clause of Part 1 is applicable.

## **12 Moisture and dust resistance**

This clause of Part 1 is applicable.

## 13 Electric strength and insulation resistance

This clause of Part 1 is applicable except as follows:

### Table 12 (13.2 of edition 3) – Insulation or disconnection test voltages

*Addition to footote p:*

In the case of **humidity sensing controls**, it may be necessary to provide specially calibrated samples to enable this test to be performed.

## 14 Heating

This clause of Part 1 is applicable.

## 15 Manufacturing deviation and drift

This clause of Part 1 is applicable except as follows:

### 15.4 Addition:

*Alternatively, the declared **manufacturing deviation** and **drift** may be expressed separately as a tolerance value to the declared **operating value**.*

### 15.5.3 Additional subclauses:

**15.5.3.101 Controls** intended for **setting by the user** shall be set at the maximum humidity value permitted by the adjustment unless otherwise declared by the manufacturer.

**15.5.3.102** The **operation** of the **control** shall be sensed by a suitable device with a sensing current not exceeding 0,05 A.

*The circuit voltage may be any convenient value that will give reliable indication of the function being monitored.*

**15.5.4** Not applicable.

## 16 Environmental stress

This clause of Part 1 is applicable.

## 17 Endurance

This clause of Part 1 is applicable except as follows:

### 17.1.3 Test sequence and conditions

*Additional subclause:*

**17.1.3.101** For **humidity sensing controls**, the tests of Clause 17 are conducted with the **activating quantity** agreed between the manufacturer and testing authority.

## **17.8 Test of automatic action at accelerated rate**

*Additional subclause:*

**17.8.4.101** *The number of automatic and manual cycles for independently mounted and in-line cord controls shall be as indicated in Clause AA.1, unless a higher number is declared by the manufacturer.*

## **17.16 Tests for particular purpose controls**

*Additional subclause and note:*

### **17.16.101 Humidity sensing controls**

- Subclauses 17.1 to 17.5 inclusive are applicable.
- 17.6 is applicable to actions classified as type 1.M or 2.M, the value of "X" being agreed between manufacturer and testing authority.
- 17.7 is applicable.
- 17.8 is applicable.
- 17.9 is applicable, except:
- 17.9.3.1 is not applicable.
- 17.10 to 17.14 inclusive are applicable.

## **18 Mechanical strength**

This clause of Part 1 is applicable.

## **19 Threaded parts and connections**

This clause of Part 1 is applicable.

## **20 Creepage distances, clearances and distances through solid insulation**

This clause of Part 1 is applicable.

## **21 Resistance to heat, fire and tracking**

This clause of Part 1 is applicable.

## **22 Resistance to corrosion**

This clause of Part 1 is applicable.

## **23 Electromagnetic compatibility (EMC) requirements – Emission**

This clause of Part 1 is applicable.

## **24 Components**

This clause of Part 1 is applicable.

## 25 Normal operation

This clause of Part 1 is applicable.

## 26 Electromagnetic compatibility (EMC) requirements – Immunity

This clause of Part 1 is applicable. See also Annex H.

## 27 Abnormal operation

This clause of Part 1 is applicable. See also Annex H.

## 28 Guidance on the use of electronic disconnection

This clause of Part 1 is applicable.

## Figures

The figures of Part 1 are applicable.

## Annexes

The annexes of Part 1 are applicable except as follows:

### **Annex H** (normative)

#### **Requirements for electronic controls**

This annex of Part 1 is applicable except as follows:

### **H.2 Terms and definitions**

#### **H.2.23 Definitions relating to functional safety**

*Additional definitions:*

##### **H.2.101**

##### **permanent operation**

continuous monitoring of the protective function during the **operation** of the appliance or system for longer than 24 h

Note 1 to entry: 24 h is considered the typical time interval between a first and a second **fault**.

##### **H.2.102**

##### **non-permanent operation**

continuous monitoring of the protective function during the **operation** of the appliance or system for less than 24 h

Note 1 to entry: 24 h is considered the typical time interval between a first and a second **fault**.

## H.6 Classification

### H.6.18 According to classes of control functions

#### H.6.18.1

*Additional note:*

NOTE 101 In general, **humidity sensing controls** perform **class A control functions**.

## H.7 Information

*Additional requirements to Table 1:*

|                                                                                                         | Information                                                                                    | Clause or subclause                                       | Method |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------|
| 58a                                                                                                     | See footnote a of Table H.101                                                                  |                                                           |        |
| 101                                                                                                     | The output condition of Type 2 <b>humidity sensing controls</b> after operation <sup>101</sup> | H.26.2.103,<br>H.26.2.104,<br>H.26.2.105                  | X      |
| 102                                                                                                     | Frequency of the <b>defined state</b> test function                                            | H.27.1.2.2.2,<br>H.27.1.2.3.2,<br>H.27.1.2.3.3            | X      |
| 103                                                                                                     | The <b>control</b> is for permanent or <b>non-permanent operation</b>                          | H.2.101.1,<br>H.2.101.2,<br>H.27.1.2.2.2,<br>H.27.1.2.3.2 | X      |
| <i>Additional footnote:</i><br><sup>101</sup> For example, conducting or non-conducting, as applicable. |                                                                                                |                                                           |        |

## H.11 Constructional requirements

### H.11.12 Controls using software

#### H.11.12.2.6 *Modification:*

*Replace the second paragraph by the following note:*

NOTE The values declared in Table 1, requirement 71, may be given in the applicable appliance standard.

#### H.11.12.2.7 *Additional note:*

NOTE 101 The values declared in Table 1, requirement 72, may be given in the applicable appliance standard.

## H.23 Electromagnetic compatibility (EMC) requirements – Emission

### H.23.1.2 Radio frequency emission

*Addition:*

Integrated and incorporated electronic **humidity sensing controls** are not subjected to the tests of H.23.1.2, as the results of these tests are influenced by the incorporation of the **humidity sensing control** into the equipment and the use of measures to control emissions used therein. They may, however, be carried out under declared conditions if so requested by the manufacturer.

## H.26 Electromagnetic compatibility (EMC) requirements – Immunity

### H.26.2 *Addition:*

After each test, one or more of the following criteria (see Subclauses H.26.2.101 to H.26.2.106) shall apply, as permitted in Table H.101.

*Additional subclauses:*

**H.26.2.101** The **control** shall remain in its current condition and thereafter shall continue to operate as declared within the limits verified in Clause 15, if applicable.

**H.26.2.102** The **control** shall assume the condition declared in Table 1, requirement 101 and thereafter shall operate as in H.26.2.101.

**H.26.2.103** The **control** shall assume the condition declared in Table 1, requirement 101, such that it cannot be **reset** automatically or manually. The output waveform shall be sinusoidal or as declared in Table 1, requirement 53 for normal **operation**.

**H.26.2.104** The **control** shall remain in the condition declared in Table 1, requirement 101. A non-self-resetting **control** shall be such that it can only **reset** manually. After the humidity which caused cut-out to occur is removed, it shall operate as in H.26.2.101 or shall remain in the declared condition as in H.26.2.103.

**H.26.2.105** The **control** may return to its initial state and thereafter shall operate as in H.26.2.101.

NOTE If a **control** is in the condition declared in Table 1, requirement 101, it may **reset** but shall resume the declared condition again if the humidity which caused it to operate is still present.

**H.26.2.106** The output and functions shall be as declared in Table 1, requirement 58a or 58b and the **control** shall comply with the requirement of 17.5.

### Table H.101 – Compliance criteria

| Applicable Clause H.26 tests                                                                                                                                                           | Compliance criteria permitted |            |            |            |            |                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------|------------|------------|------------|-------------------------|
| Type 2 <b>humidity sensing controls</b>                                                                                                                                                | H.26.2.101                    | H.26.2.102 | H.26.2.103 | H.26.2.104 | H.26.2.105 | H.26.2.106 <sup>a</sup> |
| H.26.4 to H.26.14 inclusive                                                                                                                                                            | B                             | B          | B          | A          | A          | X                       |
| Other <b>humidity sensing controls</b>                                                                                                                                                 | H.26.2.101                    | H.26.2.102 | H.26.2.103 | H.26.2.104 | H.26.2.105 | H.26.2.106 <sup>a</sup> |
| H.26.8, H.26.9                                                                                                                                                                         | X                             |            |            |            | X          | X                       |
| <p>X = Permitted</p> <p>A = Permitted when the disturbance is applied after <b>operation</b></p> <p>B = Permitted when the disturbance is applied before <b>operation</b></p>          |                               |            |            |            |            |                         |
| <p><sup>a</sup> This compliance criterion is permitted only for integrated or <b>incorporated controls</b>, since the acceptability of the output must be judged in the appliance.</p> |                               |            |            |            |            |                         |

## **H.26.5 Voltage dips, voltage interruptions and voltage variations in the power supply network**

### **H.26.5.2 Voltage variation test**

#### **H.26.5.2.2 Test procedure**

*Replacement of the last paragraph:*

The **humidity sensing control** is subjected to each of the specified voltage test cycles three times with 10 s intervals between each test cycle. For a **humidity sensing control** declared under Table 1, requirement 101, each test cycle is performed three times when the **control** is in the declared condition and three times when it is not.

## **H.26.8 Surge immunity test**

### **H.26.8.3 Test procedure**

*Additional subclause:*

**H.26.8.3.101** For **controls** declared under Table 1, requirement 101, the tests are performed when the **control** is in the declared condition and when it is not.

## **H.26.9 Electrical fast transient/burst immunity test**

### **H.26.9.3 Test procedure**

*Additional subclause:*

**H.26.9.3.101** The **control** is subjected to five tests. For **controls** declared under Table 1, requirement 101, the tests are performed when the **control** is in the declared condition and when it is not.

## **H.26.10 Ring wave immunity test**

### **H.26.10.5 Test procedure**

*Additional subclause:*

**H.26.10.5.101** For **controls** declared under Table 1, requirement 101, the tests are performed when the **control** is in the declared condition and when it is not.

## **H.26.12 Radio-frequency electromagnetic field immunity**

### **H.26.12.2 Immunity to conducted disturbances**

#### **H.26.12.2.2 Test procedure**

*Additional subclause:*

**H.26.12.2.2.101** For **controls** declared under Table 1, requirement 101, sweeping is performed when the **control** is in the declared position and when it is not.

### **H.26.12.3 Immunity to radiated disturbances**

#### **H.26.12.3.2 Test procedure**

*Additional subclause:*

**H.26.12.3.2.101** For **controls** declared under Table 1, requirement 101, sweeping is performed when the **control** is in the declared condition and when it is not.

## **H.26.13 Test of influence of supply frequency variations**

### **H.26.13.3 Test procedure**

*Additional subclause:*

**H.26.13.3.101** For **controls** declared under Table 1, requirement 101, the test shall be performed when the **control** is in the declared condition and when it is not.

## **H.26.14 Power frequency magnetic field immunity test**

### **H.26.14.3 Test procedure**

*Additional subclause:*

**H.26.14.3.101** For **controls** declared under Table 1, requirement 101, the test shall be performed when the **control** is in the declared condition and when it is not.

## **H.26.15 Evaluation of compliance**

### **H.26.15.2 Addition:**

*See Table H.101 for compliance criteria.*

### **H.26.15.4 Addition:**

*See Table H.101 for compliance criteria.*

## **H.27 Abnormal operation**

### **H.27.1.1.2 Modification:**

*Replace the first line by the following:*

*The **control** shall be operated under the following conditions. In addition, **controls** declared under Table 1, requirement 101 shall be tested when the **control** is in the declared condition and when it is not.*

### **H.27.1.1.3 Modification:**

This subclause of Part 1 is applicable except item c).

### **H.27.1.2.2 Class B control function**

This subclause of Part 1 is applicable except as follows:

#### **H.27.1.2.2.2 First fault**

*Modification:*

*Replace item b) as follows:*

- b) the **control** shall react within the **fault reaction time** (see Table 1, requirement 91) by proceeding to the **defined state** provided that a subsequent restart under the same **fault** conditions results in the **system** returning to the same **defined state** condition;

*Replace item c) as follows:*

- c) for **systems** with **non-permanent operation** only, the **control** shall continue to operate as intended, the **fault** shall be detected during the next start-up sequence. The compliance criteria shall be a) or b).

NOTE 101 Requirements for **systems** with **permanent operation** are under consideration.

*Replace item d) as follows:*

- d) the **control** shall continue to operate as intended.

*Replace the last two paragraphs with the following:*

The **fault reaction time** shall be declared by the manufacturer (see Table 1, requirement 91).

For **permanent operation** as declared by the manufacturer (see Table 1, requirement 103), item c) is under consideration.

For a **control** function, where a mechanical actuator is part of a circuit that characterizes the **defined state**, a test up to, but not including, the switching contacts is sufficient. If the test of the **defined state** fails, the **control** shall initiate the **safety shut-down**. Frequency of test is as declared by the manufacturer (see Table 1, requirement 102). Internal **faults** of the components of the checking circuits are not considered.

#### **H.27.1.2.2.3 Fault introduced during defined state**

Not applicable.

#### **H.27.1.2.3 Class C control function**

This subclause of Part 1 is applicable except as follows:

##### **H.27.1.2.3.2 First fault**

*Modification:*

*Replace item b) as follows:*

- b) the **control** reacting within the **fault reaction time** (see Table 1, requirement 91) by proceeding to **defined state** provided that subsequent restart under the same **fault** condition results in the **system** returning to the **defined state** condition;

*Replace item c) follows:*

- c) for **systems** with **non-permanent operation**, the **control** shall continue to operate as intended, the **fault** shall be detected during the next start-up sequence. The compliance criteria shall be a) or b).

NOTE Requirements for **systems** with **permanent operation** are under consideration.

*Replace item d) as follows:*

- d) The **control** shall continue to operate as intended.

*Replace the last sentence with the following:*

The **fault reaction time** shall be declared by the manufacturer (see Table 1, requirement 91).

For **permanent operation** as declared by the manufacturer (see Table 1, requirement 103), item c) is under consideration.

For a **control** function, where a mechanical actuator is part of a circuit that characterizes the **defined state**, a test up to, but not including, the switching contacts is sufficient. If the test of the **defined state** fails, the **control** shall initiate the **safety shut-down**. Frequency of test is as declared by the manufacturer (see Table 1, requirement 102). Internal **faults** of the components of the checking circuits are not considered.

#### H.27.1.2.3.3 Second fault

*Modification:*

*Replace second sentence and items a) and b) with the following:*

During assessment, for **systems** with **non-permanent operation**, the second **fault** shall only be considered to occur when a start-up sequence has been performed after the first **fault**. For **systems** with **permanent operation**, the second **fault** occurs 24 h after the first **fault**.

*Replace the last two sentences with the following:*

The **fault reaction time**, as well as the applicability of c), shall be as declared by the manufacturer.

For a **control** function, where a mechanical actuator is part of a circuit that characterizes the **defined state**, a test up to, but not including, the switching contacts is sufficient. If the test of the **defined state** fails, the **control** shall initiate the **safety shut-down**. Frequency of test is as declared by the manufacturer (see Table 1, requirement 102). Internal **faults** of the components of the checking circuits are not considered.

#### H.27.1.2.4 Faults during defined state

Under consideration.

*Additional annexes:*

## Annex AA (normative)

### Independently mounted and in-line cord controls

Table AA.1 gives the number of cycles.

**Table AA.1 – Number of cycles**

| <b>Type of control</b> | <b>Automatic action</b> | <b>Manual action</b> |
|------------------------|-------------------------|----------------------|
| Humidity sensing       | 6 000                   | 600                  |
| <b>Room humidistat</b> | 60 000                  | 600                  |

## Annex BB (normative)

### Regional differences

#### CENELEC countries

#### Annex H – Requirements for electronic controls

##### H.26.10 Ring wave immunity test

*Delete and replace by “Void”.*

#### United States

### 11 Constructional requirements

#### 11.4.5 Addition:

A capacitor may not be connected across the contacts of a **control** with a type 1.B or type 2.B action.

### 17.8 Test of automatic action at accelerated rate

#### 17.8.4.101 Addition:

The cycles are as indicated in Table BB.1.

**Table BB.1 – Minimum number of cycles for independently mounted and in-line cord controls (United States)**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Humidity sensing controls for |                          |                               |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------|-------------------------------|--------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Humidification <sup>a</sup>   |                          | Dehumidification <sup>a</sup> |                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Type 1                        | Type 2                   | Type 1                        | Type 2                   |
| Self-resetting action                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 000                         | 100 000                  | 30 000                        | 100 000                  |
| First                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 000                         | 75 000                   | 24 000                        | 75 000                   |
| Maximum cycles per minute                                                                                                                                                                                                                                                                                                                                                                                                                                   | <sub>1<sup>b</sup></sub>      | <sub>6</sub>             | <sub>6</sub>                  | <sub>6</sub>             |
| Last                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               | 25 000                   | 6 000                         | 25 000                   |
| Maximum cycles per minute                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               | <sub>1<sup>b</sup></sub> | <sub>1<sup>b</sup></sub>      | <sub>1<sup>b</sup></sub> |
| Non-self-resetting action                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6 000                         | 6 000                    | 6 000                         | 6 000                    |
| First                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 000                         | 1 000                    | 6 000                         | 1 000                    |
| Maximum cycles per minute                                                                                                                                                                                                                                                                                                                                                                                                                                   | <sub>1<sup>b</sup></sub>      | <sub>1<sup>b</sup></sub> | <sub>1<sup>b</sup></sub>      | <sub>1<sup>b</sup></sub> |
| Last                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               | 5 000                    |                               | 5 000                    |
| Maximum cycles per minute                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               | <sub>c</sub>             |                               | <sub>c</sub>             |
| Manual action                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6 000                         |                          | 6 000                         |                          |
| Maximum cycles per minute                                                                                                                                                                                                                                                                                                                                                                                                                                   | <sub>6</sub>                  |                          | <sub>6</sub>                  |                          |
| <sup>a</sup> Magnetic, manual and motor-operated switches, or the like, and switches that snap with lost motion and do not creep may be tested at the rate of six cycles per minute.<br><sup>b</sup> For all <b>controls</b> , the test is to be conducted with (50 ± 20) % ON time. A humidity operated <b>control</b> is to be so tested, using a slow rate of change.<br><sup>c</sup> Tested without current, cycle rate may be at any convenient speed. |                               |                          |                               |                          |

**17.16.101 Addition:**

If a **control** has two or more electrical ratings (for example, inductive and resistive, or different currents at different voltages), it may be tested for not less than 25 % of its declared endurance (if equal to or greater than 30 000 cycles) at each rating, but the total number of cycles on any one sample is not to be more than its declared endurance. However, at least one sample shall be tested for a total number of cycles equal to its declared endurance.

**Canada****11 Constructional requirements****11.4.5 Addition:**

A capacitor may not be connected across the contacts of a **control** with a type 1.B or type 2.B action.

**17.8 Test of automatic action at accelerated rate****17.8.4.101 Addition:**

The cycles are as indicated in Table BB.2.

**Table BB.2 – Minimum number of cycles for independently mounted and in-line cord controls (Canada)**

|                                                                                 | Humidification |                    | Dehumidification |                    |
|---------------------------------------------------------------------------------|----------------|--------------------|------------------|--------------------|
|                                                                                 | Type 1         | Type 2             | Type 1           | Type 2             |
| Self-resetting action                                                           | 6,000          | 100,000            | 30,000           | 100,000            |
| Non-self-resetting action                                                       | 6,000          | 6,000              | 6,000            | 6,000              |
|                                                                                 |                | 1,000              |                  | 1,000              |
|                                                                                 |                | 5,000 <sup>a</sup> |                  | 5,000 <sup>a</sup> |
| <sup>a</sup> Tested without current, cycle rate may be at any convenient speed. |                |                    |                  |                    |

## **Annex CC** (informative)

### **Specific regional requirements in Japan**

#### **CC.1 General**

For the purposes of this International Standard, the specific regional requirements given as follows are applicable in Japan.

##### **CC.1.1 Addition:**

NOTE 4 This regional Annex ZC shows that JIS C 9730-2-13:2010 is identical to IEC 60730-2-13/Ed.3 based on ISO/IEC Guide 21-1,8.2 Table 1, as Designation “identical” a) and b), Abbreviation “IDT” .

##### **CC.1.2 Normative references**

*Subclause 1.2 is applicable with the following modifications:*

| <b>Reference in clause 1.2</b> | <b>Replaced by</b> | <b>IDT/MOD</b> |
|--------------------------------|--------------------|----------------|
| IEC 60947                      | JIS C 8201         | MOD            |
| IEC 60730-1 Ed. 5.0:2013       | JIS C 9730-1:2016  | MOD            |

#### **CC.17.16**

This subclause is not applicable in Japan.

##### **CC.17.8.4.101**

This subclause is not applicable in Japan.

## **Annexes**

### **Annex H – Requirements for electronic controls**

#### **H.26.10 Ring wave immunity test**

This subclause is not applicable in Japan.

## Bibliography

Bibliography of Part 1 is applicable except as follows:

*Addition:*

ISO/IEC Guide 21-1:2005, *Regional or national adoption of International Standards and other International Deliverables – Part 1: Adoption of International Standards*

JIS C 9730-2-13:2010, *Automatic electrical controls for household and similar use – Part 2-13: Particular requirements for humidity sensing controls*

---



INTERNATIONAL  
ELECTROTECHNICAL  
COMMISSION

3, rue de Varembé  
PO Box 131  
CH-1211 Geneva 20  
Switzerland

Tel: + 41 22 919 02 11  
Fax: + 41 22 919 03 00  
[info@iec.ch](mailto:info@iec.ch)  
[www.iec.ch](http://www.iec.ch)